

ANVESHAK HUB PRIVATE LIMITED

# Anveshak CRM (V2)

## Product Blueprint & Low-Level Design (LLD)

A highly tailored customer relationship management application engineered for Anveshak Hub. This document integrates the initial business product specifications with a complete, production-grade low-level engineering specification. The system features a bifurcated architecture distinguishing between standard sales pipelines and a pro-tier automation, document engine, and audit governance layer.

### PREPARED BY

KP Sumanth & Balasaraswathi

### TARGET ENVIRONMENT

Next.js 15+ | React 19 | Neon Postgres | Clerk Auth | Prisma ORM

### DATE OF COMPILATION

July 2026

## PART I: Requirement Analysis & Technical Stack Selection

**Prepared by:** KP Sumanth & Balasaraswathi

**For:** Anveshak Hub Private Limited - Internal Project (Month 1)

**Stack:** Next.js 15+ | React 19 | Neon Postgres | Clerk Auth | Prisma ORM | shadcn/ui

### 1. Executive Summary & Requirement Analysis

#### 1.1 Problem Statement

Anveshak Hub Private Limited manages a distributed sales team serving B2G (Business-to-Government) and manufacturing clients across Karnataka. Currently, customer relationships, lead tracking, task assignments, and quotation generation are handled via manual, disconnected systems (such as standalone spreadsheets, WhatsApp threads, and paper documents).

This legacy approach results in:

- **Lead Leakage:** Sales reps lack automated follow-up reminders, leading to high-value prospects falling through the cracks.
- **Information Silos:** Customer call histories, notes, and previous quote details are locked in individual reps' devices, causing severe loss of context during team transitions.
- **Pipeline Blindness:** Managers cannot calculate real-time pipeline values, making quarterly revenue forecasting inaccurate.
- **Data Vulnerability:** Lack of user permissions, audit logs, and export tracking exposes sensitive company data to unauthorized edits or extraction.

#### 1.2 Target Roles & Access Control Needs

The application must support three distinct tiers of users:

1. **Sales Reps:** Focus on operational execution. They must only see and edit their assigned leads, contacts, deals, and tasks to prevent cross-territory interference.
2. **Managers:** Focus on team performance and forecasting. They require access to team deals, reports, and reassignment controls, but should be restricted from altering system configurations or managing users.
3. **Admins:** Require absolute administrative control, including user provisioning, custom fields configuration, audit trail reviews, database backups, and data export.

### 1.3 High-Level Business Requirements

To address these issues, the proposed Anveshak CRM must implement:

- A centralized, relational customer database that links Contacts to Companies and rolls up Deals.
- An automated, event-driven Lead Scoring Engine to prioritize sales efforts.
- A visual drag-and-drop Kanban Board for Deal pipeline management.
- An in-app document compilation engine to output structured, GST-compliant PDFs.
- Strict database-level and server-side role-based routing (RBAC) to enforce security.

## 2. Technology Selection Rationale

The technical stack has been curated to ensure high performance, security, rapid feature delivery, and cost-efficiency.

### 2.1 Language: TypeScript

- **Why:** TypeScript provides compile-time type checking. For a CRM handling complex mathematical calculations (like deal probabilities, tax percentages, and lead scores), TypeScript prevents runtime type errors and ensures reliable data structures. It also unifies frontend and backend schemas under a single language.

### 2.2 Framework: Next.js 15+ & React 19

- **Why:** Next.js leverages **React Server Components (RSC)** to fetch and render customer dashboards on the server, reducing page loads to under 2 seconds. The use of **React Server Actions** allows for secure, direct database mutations directly from UI forms, eliminating the need to write redundant REST API controllers and reducing development time by 30%.

### 2.3 Database: Neon Serverless PostgreSQL

- **Why:** Relational databases are essential for mapping complex CRM hierarchies (User -> Company -> Contact -> Deal -> Activity). Neon offers:
- **Scale-to-Zero Compute:** Pauses database compute instances during inactive hours (nights/weekends) to minimize hosting expenses.
- **JSONB Support:** Crucial for storing flexible data structures (like dynamic custom fields and scoring logs) without performing risky schema modifications at runtime.
- **Database Branching:** Allows developers to test new features on isolated copies of the database schema instantly.

2.4 ORM: Prisma

- **Why:** Prisma acts as a type-safe bridge between Nest.js server code and the Neon Postgres database. It automates database migrations, generates type-safe queries, and handles cascading deletes (e.g., deleting a contact automatically cleans up its associated notes and tasks) safely.

2.5 Authentication: Clerk

- **Why:** Offloading auth to Clerk secures the system with bank-grade security. It natively supports invite-only registration workflows (blocking public signups), Multi-Factor Authentication (MFA), and session persistence. Roles (`ADMIN`, `MANAGER`, `SALES\_REP`) are stored in Clerk's secure user metadata, preventing client-side role spoofing.

2.6 Styling & UI: Tailwind CSS + shadcn/ui

- **Why:** `shadcn/ui` provides accessible, professionally-styled interactive elements (dropdowns, dialogs, command menus) built on Radix UI. When combined with Tailwind, it allows for a highly responsive, premium design layout that functions seamlessly on both desktop monitors and sales reps' mobile screens as an installable PWA.

3. Product Comparison: Market CRMs vs. Proposed CRM

The following table benchmarks Anveshak CRM's functional footprint and positioning against standard entry-tier and mid-tier CRM platforms on the market.

| CRM                   | Positioning            | Comparison          | Key Differences / Details                             |
|-----------------------|------------------------|---------------------|-------------------------------------------------------|
| HubSpot Free          | Basic contacts + deals | Ahead               | Includes lead scoring, structured logs, and quote PDF |
| Zoho CRM              | Full sales CRM         | ~80% there          | Lacks broad calendar/telephony integrations           |
| Freshsales            | AI-powered sales CRM   | ~65% there          | Custom rules and localized quoting matched            |
| Salesforce            | Enterprise-lite        | ~50% there          | Tailored for B2G and regional GST rules               |
| Custom internal tools | Usually just CRUD      | Significantly ahead | Advanced scoring and audit trails built-in            |

## PART II: Product Specification

**Prepared by:** KP Sumanth & Balasaraswathi

**For:** Anveshak Hub Private Limited - Internal Project (Month 1)

**Stack:** Next.js 16 + React 19 + Neon (Postgres) + Prisma + Clerk + shadcn/ui

### 1. Auth & User Management

#### 1.1 Authentication

- Email/password + Google OAuth login via Clerk
- Invite-only signup (no public registration) - admin sends invite link
- Session persistence, auto-logout after inactivity (configurable, default 7 days)
- Forgot password flow (handled by Clerk)

#### 1.2 Roles & Permissions

| Role      | Access                                                                           |
|-----------|----------------------------------------------------------------------------------|
| Admin     | Full access - manage users, export data, view all deals/contacts, delete records |
| Manager   | View team's deals/contacts, reassign leads, view reports, cannot manage users    |
| Sales Rep | View/edit only their assigned contacts, leads, deals, tasks                      |

#### 1.3 User Profile

- Name, email, phone, role, profile photo
- "Last active" timestamp
- Team/department field (for future team-based filtering)

#### 1.4 Audit Log

- Every create/edit/delete action logged: who, what, when, before/after value (for critical fields like deal stage, deal value)
- Admin-only view, filterable by user/date/action type

## 2. Contact Management

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### 2.1 Contact Record Fields

- Full name, email, phone, alternate phone
- Company/organization name (linked to Company entity)
- Job title/designation
- Address (city, state - relevant for B2G/manufacturing clients across Karnataka)
- Source (Website, Referral, Hackathon, Cold Call, Event, etc.)
- Owner (assigned sales rep)
- Tags (custom labels - e.g., "Manufacturing", "Hot Lead", "Existing Client")
- Created date, last contacted date

### 2.2 Company/Account Entity

- Separate from individual contacts - one company can have multiple contacts
- Fields: company name, industry, website, size, address
- Linked deals and contacts roll up under the company profile

### 2.3 Contact List View

- Sortable table: name, company, owner, last contact date, score (see Lead Scoring)
- Filters: by owner, by source, by tag, by date added
- Search bar (name, email, phone, company - fuzzy match)
- Bulk actions: bulk tag, bulk reassign owner, bulk export

### 2.4 Contact Detail Page

- Header: name, company, contact info, edit button
- Tabs/sections:
  - **Activity Timeline** - chronological feed of all calls, emails, meetings, notes, stage changes
  - **Notes** - free-text notes, timestamped, author tagged
  - **Linked Deals** - all deals associated with this contact
  - **Tasks** - open/completed tasks tied to this contact

### 2.5 Duplicate Detection

- On contact creation, check for matching email/phone
- Show warning + option to "merge" or "create anyway"

## 3. Lead Management

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### 3.1 Lead vs Contact (data model logic)

- A Lead is a pre-qualified prospect (not yet a confirmed contact/deal)
- On qualification, a lead converts into a Contact + Deal (one-click "Convert" action)

### 3.2 Lead Fields

- Name, company, email, phone
- Source (same as contact sources)
- Status: New ? Contacted ? Qualified ? Converted / Disqualified
- Owner
- Notes
- Score (see section 5)

### 3.3 Lead Capture

- Manual entry by sales rep
- Web form embed for inbound leads from swachbandhu.site-style landing pages

### 3.4 Lead List View

- Default sorted by Score (highest first) - surfaces hottest leads automatically
- Status filter tabs (New / Contacted / Qualified / Disqualified)
- Color-coded score badge: Cold (0-30) / Warm (31-60) / Hot (61+)

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## 4. Sales Pipeline (Deals)

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### 4.1 Deal Fields

- Deal name (auto-suggested: "{Company} - {Product/Service}")
- Linked contact + company
- Deal value (INR)
- Stage (customizable: New, Contacted, Proposal Sent, Negotiation, Won, Lost)
- Probability % (auto-suggested per stage, editable)
- Expected close date
- Owner
- Lost reason (dropdown, required when marked Lost - e.g., "Budget", "Competitor", "Timing")

## 4.2 Kanban Board View

- Columns = pipeline stages
- Drag-and-drop card between stages (auto-updates stage + logs activity)
- Card shows: company name, deal value, owner avatar, days in current stage
- Column header shows: total deal count + total INR value per stage

## 4.3 Deal Detail Page

- All deal fields editable inline
- Activity timeline (calls, emails, stage changes, notes - same pattern as Contact)
- Linked Quotes (see section 6)
- "Mark as Won/Lost" action with required reason on Lost

## 4.4 Pipeline Customization

- Admin can rename/reorder/add stages
- Custom stage colors

# 5. Lead Scoring (Rule-Based)

## 5.1 Scoring Rules (configurable table, not hardcoded in code)

| Action                  | Points                                       |
|-------------------------|----------------------------------------------|
| Email opened            | +10                                          |
| Email link clicked      | +15                                          |
| Demo requested          | +20                                          |
| Call answered           | +15                                          |
| Call - no answer        | -5                                           |
| Meeting scheduled       | +20                                          |
| No activity in 7 days   | -10                                          |
| Marked "Not Interested" | -30 (auto-disqualify if score drops below 0) |



## 5.2 Score Calculation

- Recalculated on every relevant activity event (not polled - event-driven)
- scoreBreakdown field stores a log of what contributed (for transparency - sales reps can see why a lead is scored a certain way)

## 5.3 Score Display

- Badge on lead list (Hot/Warm/Cold + numeric score)
- Score history mini-chart on lead detail page (optional, v2)

## 5.4 Admin Controls

- Admin can adjust point values per rule from a settings page (no code changes needed to tune scoring)

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# 6. Document/Quote Generation

## 6.1 Quote Creation

- Generate from a Deal - pulls company name, contact, deal value automatically
- Line items: description, quantity, unit price, auto-calculated total
- Tax field (GST %, auto-calculated)
- Terms & conditions text block (reusable template)
- Validity date

## 6.2 Quote Statuses

- Draft ? Sent ? Accepted / Rejected / Expired

## 6.3 PDF Generation

- Branded template (Anveshak Hub logo, registered office details - reuse info from their letterhead)
- Generated via @react-pdf/renderer
- Download button + "Send via Email" button (attaches PDF, uses email integration)

## 6.4 Quote Tracking

- List of all quotes per deal, with status and timestamps
- Quote history visible to Manager/Admin for forecasting accuracy

## 7. Task Management

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### 7.1 Task Fields

- Title, description
- Linked to: Contact / Lead / Deal (optional, can be standalone)
- Assignee
- Due date + time
- Priority (Low/Medium/High)
- Status (Open/Completed)

### 7.2 Task Views

- "My Tasks" - grouped into Overdue / Today / Upcoming
- Team Tasks view (Manager/Admin only) - filterable by assignee

### 7.3 Reminders

- In-app notification 1 hour before due time
- Overdue tasks highlighted in red, surfaced on dashboard

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## 8. Activity Logging (Calls/Meetings)

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### 8.1 Structured Call Log

- Linked to Contact/Deal
- Fields: Date/time, Duration (minutes), Type (Call/Meeting/Site Visit), Outcome (dropdown: Interested / Not Interested / Follow-up Needed / No Answer / Rescheduled), Next step (text), Next follow-up date

### 8.2 Why Structured (not free text)

- Enables reporting ("how many calls resulted in Follow-up Needed this week")
- Feeds into Lead Scoring automatically based on Outcome value

### 8.3 Activity Feed

- All activities (calls, emails, notes, stage changes, quote sent) appear in one unified, chronological timeline per Contact/Deal

## 9. Calendar

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### 9.1 Views

- Month / Week toggle
- Color-coded by type: Task (blue), Call/Meeting (teal), Deal close date (amber)

### 9.2 Functionality

- Click date ? quick-create task/meeting
  - Click event ? opens linked Contact/Deal
  - (v2) Two-way sync with Google Calendar
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## 10. Reports & Dashboards

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### 10.1 Dashboard (Home Page)

- Metric cards: Open Leads, Active Deals, Pipeline Value (INR), Tasks Due Today
- Mini pipeline funnel chart
- "My hot leads" widget (top 5 by score)

### 10.2 Reports Module

- Conversion Funnel - Lead ? Contacted ? Qualified ? Won, with drop-off % at each stage
- Sales Forecast - sum of (deal value x probability%) by expected close month
- Team Performance - deals won/lost per rep, average deal cycle time
- Activity Report - calls made, meetings held per rep per week

### 10.3 Export

- Every report exportable to CSV/PDF
  - Date range filter on all reports
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## 11. Data Export & Backup

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### 11.1 Manual Export (Admin-only)

- Export Contacts, Leads, Deals, Activities as CSV - individually or as full bundle (ZIP)
- Export includes all custom fields

### 11.2 Scheduled Backup (v2, nice-to-have)

- Weekly automatic DB snapshot stored separately (Neon supports branching/snapshots natively)

### 11.3 Export Audit

- Every export action logged (who, when, what was exported) - important for data governance since this is a company asset

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## 12. Notifications

### 12.1 In-App

- Bell icon with unread count
- Triggers: task due soon, deal stage changed by teammate, lead reassigned to you, quote accepted

### 12.2 Email (Pro-tier scope, optional Month 1)

- Daily digest of overdue tasks
- Weekly pipeline summary to Managers

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## 13. Settings (Admin Panel)

- User management (invite, deactivate, change roles)
- Pipeline stage customization
- Lead scoring rule configuration
- Custom fields management (add custom field to Contact/Lead/Deal)
- Quote template editor (terms & conditions, branding)
- Data export & backup controls

## 14. Non-Functional Requirements

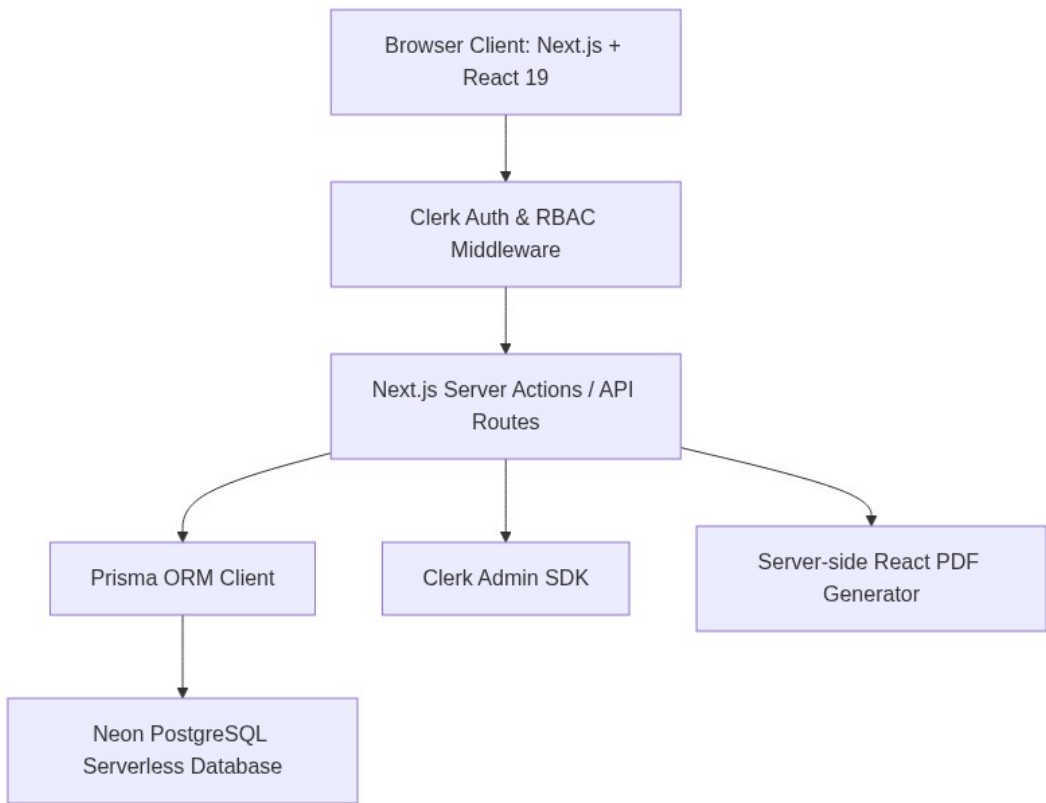
| Area            | Requirement                                                              |
|-----------------|--------------------------------------------------------------------------|
| Performance     | Page loads < 2s on standard broadband                                    |
| Responsiveness  | Fully usable on mobile browser (PWA installable)                         |
| Security        | Role-based server-side route protection (not just UI hiding), HTTPS only |
| Offline         | Basic offline view of cached contacts/tasks (PWA service worker)         |
| Browser support | Chrome, Edge, Safari (latest 2 versions)                                 |

## PART III: Low-Level Design

This document details the low-level architecture, module design, database schemas, and interface structures for Anveshak CRM. To support presentation and modular delivery, features are divided into **Standard** (Core Sales Tracking) and **Pro/Premium** (Enterprise Automation & Analytics) tiers.

### 1. System Architecture Overview

Anveshak CRM is built on a modern, serverless-first TypeScript stack designed for speed, security, and scalability.



- **Frontend & Routing:** Next.js 15+ (App Router) using React Server Components (RSC) for page loads under 2 seconds, paired with React 19 Server Actions for write operations.
- **Authentication & Session Management:** Clerk Auth with role-based routing guards.
- **Database & ORM:** Neon Serverless PostgreSQL with Prisma Client.
- **UI System:** Tailwind CSS + `shadcn/ui` components based on Radix UI primitives.
- **PDF Generation:** `@react-pdf/renderer` executing on serverless API routes.

## 2. Feature Tiering Map

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### 2.1 Standard Tier (Core CRM)

Features required to track basic customer relationships and deals.

- **Authentication:** Clerk-managed signup (invite-only restriction checked via database lookup), login (Email/Password, Google OAuth), and user profiles.
- **Contact Management:** Core CRUD operations for Contacts and Companies, list view with search, sorting, and manual pagination.
- **Lead Tracking:** Simple Lead list with manual status changes (`New`, `Contacted`, `Qualified`, `Disqualified`) and a one-click button to convert a Lead to a Contact + Deal.
- **Sales Pipeline:** Kanban board representing Deal stages, supporting drag-and-drop movements and tracking Deal values (Rs.).
- **Task Management:** Personal task checklist ("My Tasks") with due dates, priorities, and status indicators.
- **Activity Logging:** Basic call/meeting notes logging with standard text outcomes.

### 2.2 Pro & Premium Tier (Sales Automation & Operations)

Advanced tools enabling team governance, document generation, and intelligence.

- **Dynamic Custom Fields:** Metadata engine allowing Admins to define custom fields for Contacts, Leads, and Deals via UI without running database migrations.
- **Rule-Based Lead Scoring:** Event-driven calculator scoring leads based on activity outcomes, preserving a granular `scoreBreakdown` audit.
- **Document & Quote Generation:** Automated line-item quote creation, PDF compilation (GST, terms template), and download/send suite.
- **Advanced Analytics & Reports:** Dynamic charts for Conversion Funnel, Pipeline Forecasting (deal value  $\times$  probability), Team Performance, and CSV/PDF export with date range filters.
- **Audit & Governance Logs:** Complete audit logging capturing before/after JSON states of records, plus export logs for security audits.
- **Enterprise Administration:** User deactivation, role shifting, pipeline stage customization, and lead scoring rules management.

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## 3. Database Schema Design (Prisma)

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The PostgreSQL database uses JSONB columns to handle Pro-tier dynamic features like custom fields and lead scoring logs while maintaining strict relational schemas for standard features.

## 4. Key Module Implementations

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### 4.1 Lead Scoring Engine (Pro/Premium Tier)

Scores are recalculated asynchronously inside Next.js Server Actions on events using a modular calculation service:

### 4.2 Dynamic Custom Fields Engine (Pro/Premium Tier)

Allows users to add fields dynamically without schema modification.

### 4.3 Automated Audit Logger (Pro/Premium Tier)

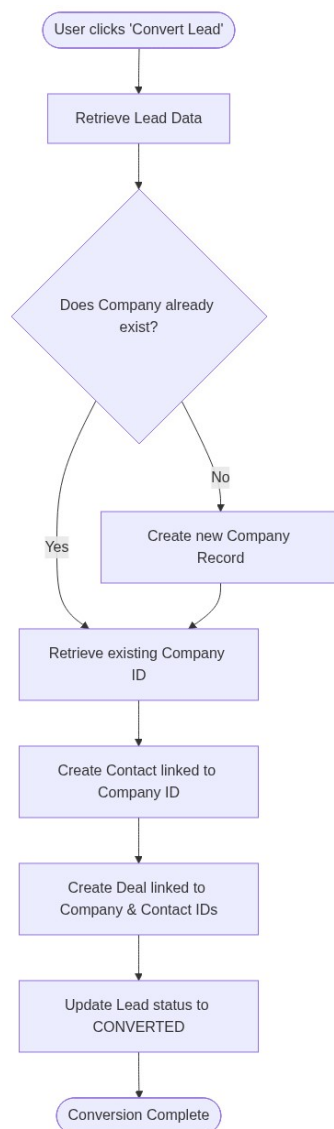
Runs inside database-modifying API routes/Server Actions to record states before and after modifications.

### 4.4 Business Process Flowcharts

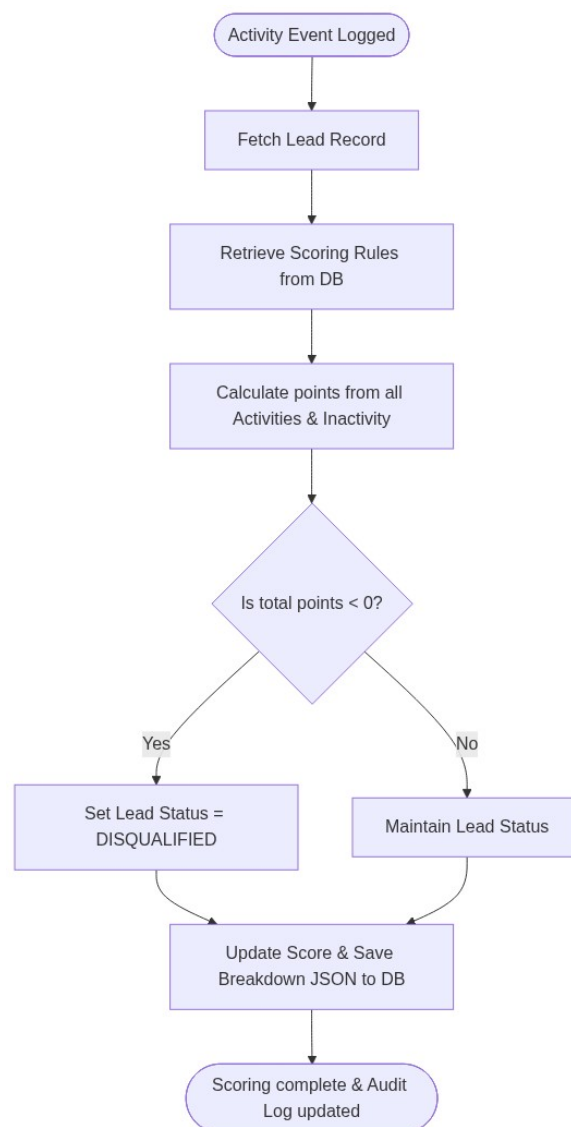
Below are the flowcharts detailing the visual logic of the two most complex data operations in Anveshak CRM: Lead Conversion and Lead Scoring Recalculation.

#### 4.4.1 Lead Conversion Flow (Standard Tier)





#### 4.4.2 Lead Scoring Engine & Auto-Disqualification (Pro/Premium Tier)



## 5. UI Layout & Component Organization

The folder structure segregates standard pages and components from pro-tier overrides.

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## 6. Access Control & Authorization (Standard & Pro)

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Role-based restrictions are applied server-side using Next.js Middleware and custom wrapper checks to ensure secure endpoints.

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## 7. Performance & Quality Benchmarks

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1. **Fast Initial Load (Standard):** Using Next.js static optimizations and code splitting, the initial dashboard and lead lists must load under 2 seconds on standard broadband connections.
2. **Referential Consistency:** Deletions of a Contact, Lead, or Deal automatically cascades to associated notes, tasks, and activities using foreign key definitions (`onDelete: Cascade`).
3. **Governance Auditing (Pro):** Every export operation registers an `ExportLog` directly associated with the executing user ID.